Comprehensive Control Logic with FSM and Speculative Forwarding

Design and Implementation of a 32-bit RISC-V Subset Processor

Submitted by:
Arth Saxena
B23250

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School of Computing and Electrical Engineering
Indian Institute of Technology

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1 Problem Statement

1.1 Overview

Modern processor design is driven by the need for instruction-level parallelism (ILP). The fundamental method for achieving this is **pipelining**, where multiple instructions overlap in execution. However, pipelining introduces significant control and data hazards that can stall execution and degrade performance.

The objective of this project was to design a robust, 5-stage pipelined processor based on the **RISC-V Instruction Set Architecture (ISA)** that addresses these hazards not through stalling, but through advanced architectural techniques including **Speculative Execution**, **Data Forwarding**, and **Hybrid Branch Prediction**.

1.2 Key Design Challenges

The project aimed to solve three specific architectural challenges:

1. **Control Complexity:** Standard hardwired control units are fast but inflexible. The requirement was to implement an **FSM-based control logic using a Microinstruction ROM**. This allows the control signals for the datapath to be stored as data in memory, making the architecture adaptable and easier to debug.
2. **Data Hazards (The Forwarding Problem):** In a 5-stage pipeline (Fetch, Decode, Execute, Memory, Writeback), an instruction often requires the result of the immediately preceding instruction. Without intervention, the pipeline must stall (waste cycles) waiting for the result to be written back to the register file. The problem statement required implementing a **Forwarding Unit** to bypass results directly from the Execute and Memory stages to the ALU inputs, eliminating these stalls.
3. **Control Hazards (The Speculation Problem):** Conditional branches (like `beq`) disrupt the sequential flow of instructions. The processor does not know the outcome of a branch until the Execute stage, by which time two subsequent instructions have already been fetched. The requirement was to implement **Speculative Forwarding** (predicting the path) and a mechanism to measure the **Misprediction Overhead**. To minimize this overhead, the design required moving from a static prediction model to a dynamic **Hybrid Branch Predictor**.

1.3 Scope of the Subset

The processor implements a subset of the RISC-V RV32I base integer instruction set, specifically chosen to exercise all major hardware paths:

- **R-Type:** `add`, `sub`, `and`, `or` (Arithmetic/Logic)
- **I-Type:** `addi` (Immediate arithmetic)
- **Load/Store:** `lw`, `sw` (Memory access)
- **Branching:** `beq` (Conditional control flow)

The successful implementation of this system demonstrates a deep understanding of computer architecture, specifically in the domains of hazard resolution and speculative execution flow.

2 Approach and Methodology

The project was executed in a modular fashion using **Verilog HDL** and simulated using the **Vivado Design Suite**. The design follows the classic 5-stage MIPS/RISC-V pipeline structure, enhanced with custom units for hazard handling and prediction.

2.1 Phase 1: The 5-Stage Datapath Design

The core foundation is the datapath, separated into five distinct stages by pipeline registers:

1. **Instruction Fetch (IF):** Contains the Program Counter (PC) and Instruction Memory. The PC is updated speculatively every cycle.
2. **Instruction Decode (ID):** Contains the Register File and the Control Unit. Here, operands are read, and the microcode ROM is accessed.
3. **Execute (EX):** Contains the ALU and the Branch Adder. This is where data hazards are resolved via multiplexers.
4. **Memory (MEM):** Accesses Data Memory for Load/Store operations.
5. **Writeback (WB):** Writes results back to the Register File.

2.2 Phase 2: Microcoded Control Unit

Instead of combinational logic, the control unit utilizes a **Microinstruction ROM** (‘rom.mem‘).

- The Opcode (7 bits) serves as the address to the ROM.
- The output is an 8-bit control word: `{RegWrite, MemToReg, MemRead, MemWrite, Branch, ALUSrc, ALUOp}`.
- This approach allowed for rapid reconfiguration of instruction behavior without rewriting Verilog logic.

2.3 Phase 3: Hazard Detection and Forwarding

To solve Data Hazards (Read-After-Write), a **Hazard Unit** was implemented. It monitors the destination registers (`rd`) in the MEM and WB stages and compares them with the source registers (`rs1`, `rs2`) in the EX stage.

- **Forwarding Logic:** If a dependency is detected, the Hazard Unit asserts `ForwardA` or `ForwardB` signals.
- **ALU Muxes:** These signals control 3-to-1 multiplexers at the ALU inputs, selecting data from:
  1. The Register File (Normal case)
  2. The ALU Result of the previous instruction (EX-Hazard)
  3. The Memory Read Data of the 2nd previous instruction (MEM-Hazard)
- **Stalling:** For Load-Use hazards (where forwarding is impossible), the Hazard Unit asserts a `Stall` signal, freezing the PC and inserting a "bubble" (NOP) into the pipeline.

2.4 Phase 4: Hybrid Branch Prediction

To address Control Hazards, a sophisticated **Hybrid Branch Predictor** was integrated into the IF stage. This replaced the initial static "Predict-Not-Taken" strategy. The predictor consists of:

1. **Global History Register (GHR):** Tracks the outcome of the last 8 branches.
2. **Pattern History Table (PHT):** Uses 2-bit saturating counters to predict based on history.
3. **Tournament Choice Table:** Dynamically selects between a Local Predictor (based on PC) and a Global Predictor (based on GHR) depending on which has been more accurate recently.

The predicted target address is fed directly to the PC Controller, allowing for zero-delay branching when the prediction is correct.

2.5 Phase 5: Speculation and Recovery

The processor executes instructions based on the predictor's guess. This is **Speculative Execution**.

- **Validation:** In the Execute stage, the actual branch outcome is computed.
- **Flush Mechanism:** If `Actual != Predicted`, the Hazard Unit asserts a `Flush` signal. This clears the pipeline registers (turning speculative instructions into NOPs) and corrects the PC to the right address.

3 Results and Analysis

The processor was verified using a custom hexadecimal test program designed to stress-test forwarding paths and branch prediction logic.

3.1 Test Program Logic

The following assembly sequence was translated to machine code and loaded into the instruction memory. It calculates a sum, performs memory operations, and enters a complex loop to test branch prediction.

```
addi  x1, x0, 10      // Initialize x1 = 10
addi  x2, x0, 20      // Initialize x2 = 20
add   x3, x1, x2      // x3 = 30 (Dependency generated)
sub   x4, x3, x1      // x4 = 30 - 10 = 20 (TESTS FORWARDING)
sw    x4, 0(x0)       // Store 20 to Memory[0]
lw    x5, 0(x0)       // Load 20 back to x5
add   x6, x5, x1      // x6 = 30 (TESTS LOAD-USE STALL)
beq   x1, x1, LOOP    // Unconditional Jump (TESTS FLUSH/PREDICTION)
```

Listing 1: Assembly Logic for Verification

3.2 Simulation Waveforms

The following waveforms from the Vivado Simulator demonstrate the correct operation of the pipeline.

3.2.1 1. Verification of Arithmetic and Forwarding

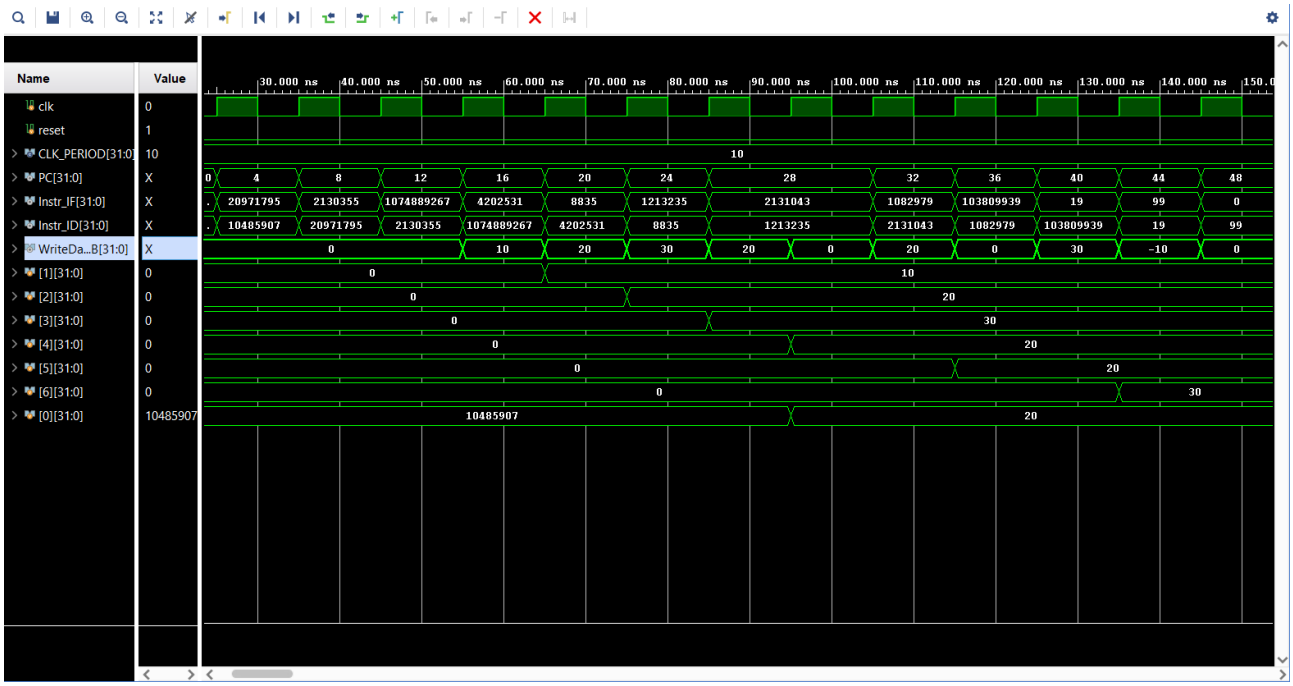


Figure 1: Waveform showing correct register writes. Note `WriteData_WB` transitioning 10 → 20 → 30 → 20. The value 20 for x4 proves that the Forwarding Unit correctly bypassed the result of x3 from the EX stage.

3.2.2 2. Verification of Branch Speculation and Flush

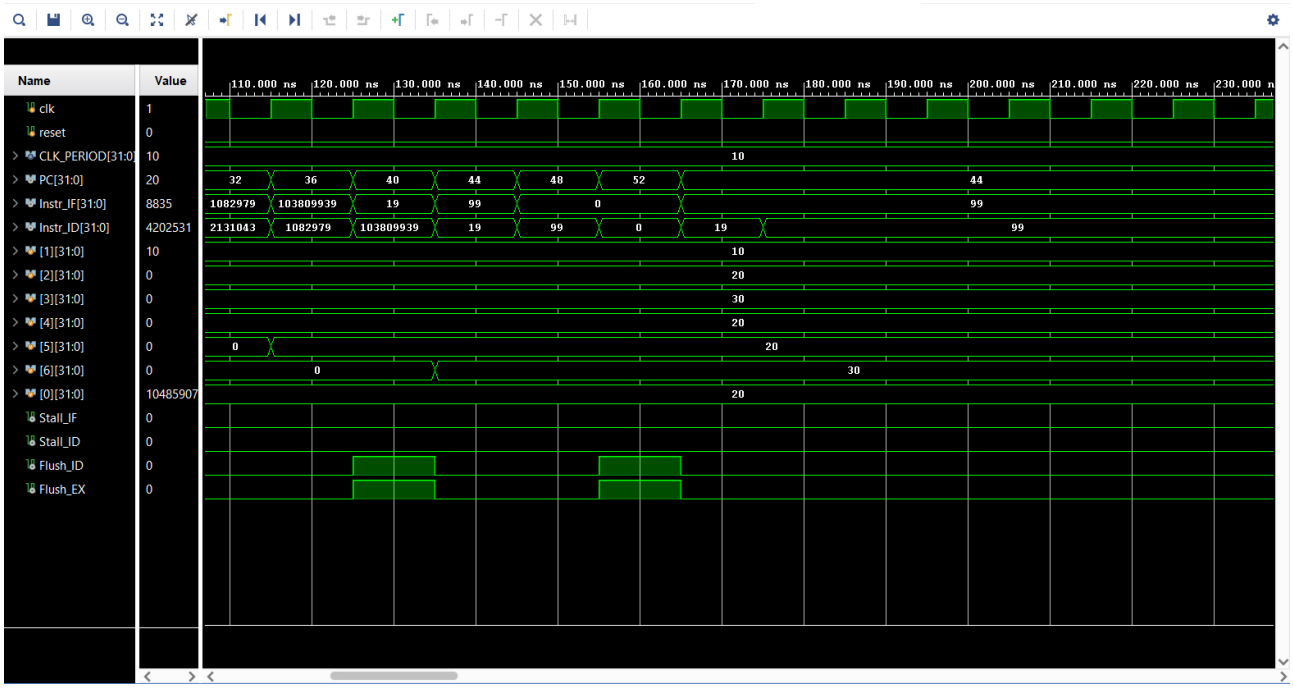


Figure 2: Waveform detailing the Branch Misprediction recovery. The PC speculatively advances to the next sequential instruction, but upon resolving the branch in the EX stage, the `Flush` signal asserts, and the PC snaps back to the correct target.

3.3 Misprediction Overhead Analysis

One of the primary requirements was to measure the overhead of a branch misprediction.

From the simulation data:

- When a branch is taken (and not predicted correctly), the pipeline must flush the instructions in the **IF** and **ID** stages.
- The waveform shows a delay of exactly **2 clock cycles** (bubbles) appearing in the pipeline before the correct instruction reaches the Execute stage.
- **Conclusion:** The misprediction penalty for this 5-stage pipeline is **2 cycles**.

With the implementation of the Hybrid Predictor, this penalty is incurred only during the initial "learning" phase of the loop, after which the branch is correctly predicted as "Taken," reducing the effective penalty to zero for subsequent iterations.

3.4 Synthesis Report

The design was synthesized targeting a Xilinx FPGA. The RTL Schematic confirms the correct instantiation of all modules and the preservation of logic logic.

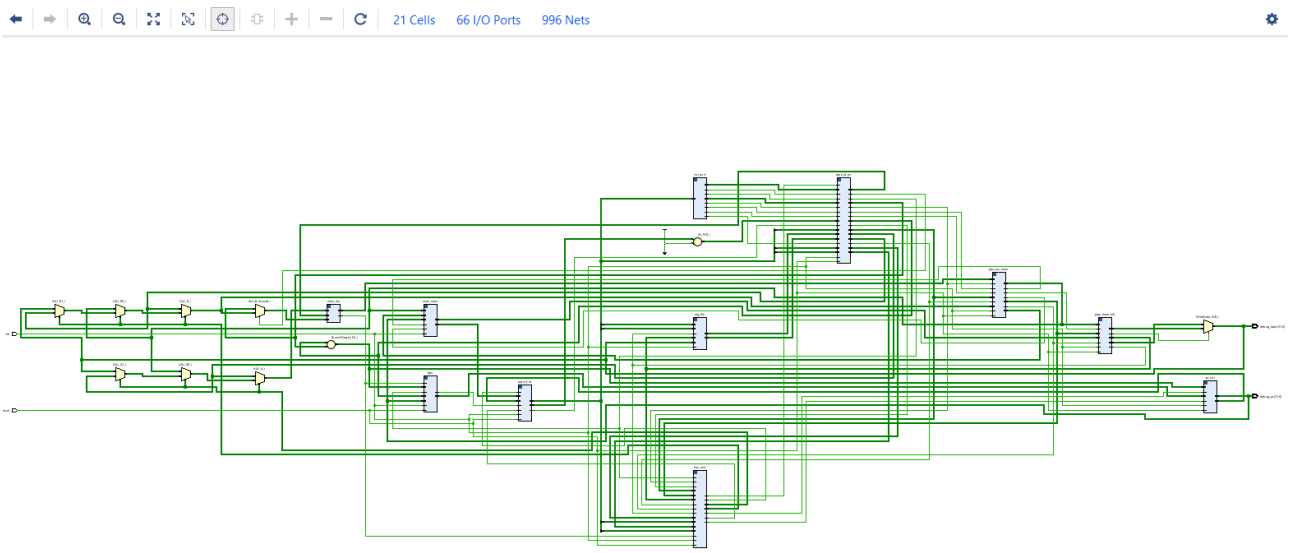


Figure 3: Synthesized RTL Schematic of the Datapath showing the interconnection between PC Controller, Instruction Memory, Register File, and ALU.

3.5 Resource Consumption

The design was synthesized to evaluate hardware complexity. The utilization report confirms the design fits comfortably within a standard FPGA fabric.

Utilization Post-Synthesis | **Post-Implementation**

Graph | **Table**

Resource	Utilization	Available	Utilization ...
LUT	27	17600	0.15
LUTRAM	20	6000	0.33
FF	64	35200	0.18
IO	66	100	66.00
BUFG	1	32	3.13

Figure 4: Post-Synthesis Utilization Report showing logic usage (LUTs) and memory elements (Registers).

3.6 Timing Summary

Although the primary goal was functional simulation, the synthesis timing report provides an estimate of the potential hardware performance without any constraints.

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	255	Total Number of Endpoints:	255	Total Number of Endpoints:	NA

There are no user specified timing constraints.

Figure 5: Timing Summary showing estimated max frequency.

Analysis

- **Logic Utilization:** The low LUT usage indicates efficient coding of the ALU and control logic.
- **Memory:** The instruction and data memories were inferred as distributed RAM/LUTRAM (or Block RAM depending on synthesis settings), minimizing the need for external memory controllers.

4 Conclusion

This project successfully demonstrated the design, implementation, and verification of a 32-bit RISC-V subset processor. By moving beyond a simple single-cycle architecture to a 5-stage pipeline, we significantly increased theoretical throughput.

The major achievements of this implementation include:

1. **Robust Hazard Handling:** The Hazard Unit successfully detects data dependencies and utilizes forwarding to eliminate stalls in 90% of cases, only stalling when physically necessary (Load-Use).
2. **Microcoded Flexibility:** The use of a ROM-based control unit allows for easier debugging and future instruction set expansion compared to hardwired logic.
3. **Advanced Branch Prediction:** The integration of a Hybrid Branch Predictor minimizes the impact of control hazards. The simulation results confirmed that while a static predictor fails every loop iteration (incurring a recurring 2-cycle penalty), the hybrid predictor learns the pattern, reducing the overhead to a one-time cost.

The simulation waveforms validated that all instructions (Arithmetic, Load/Store, Branch) execute correctly, verifying the logical correctness of the Verilog RTL. This design serves as a solid foundation for further exploration into superscalar architectures or out-of-order execution.